

L11b: Covered Calls and Protective Collars

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Objectives for Today

Today, we analyze two strategies that combine shares with options. What can each trade earn and lose, and how does its value change with the share price?

Today's concepts

- Covered calls: shares plus a short call, two regimes, breakeven, maximum profit and loss.
- Protective collars: shares plus a long put and a short call, three regimes, breakeven, maximum profit and loss.
- Net Delta: how the short call and the long put lower the Delta of the shares.

What You Need to Know

Core concepts

- Covered call profit per share: $\min(S(T), K) - \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$. The premium lowers the breakeven, and the call caps the profit at the strike.
- Collar profit per share, put at K_1, call at K_2: $S(T) + (K_1 - S(T))^+ - (S(T) - K_2)^+ - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$. The put sets a floor, and the call a cap.
- Net Delta: $mn(1 - \Delta_c)$ for a covered call and $mn(1 + \Delta_p - \Delta_c)$ for a collar. A collar's put and call can only lower its net Delta from that of the shares.

For understanding (you won't reproduce these)

- The derivation of the net Delta of a portfolio, delta-neutral positions, and the composite Delta table (derivation notebook).

Lectures and Examples

Lecture:

CHEME-5660-L11b-Lecture-CoveredCalls-Collars-Fall-2026.ipynb

Examples:

CHEME-5660-L11b-Example-AnalysisCoveredCall-Fall-2026.ipynb

CHEME-5660-L11b-Example-StandardListed-SPY-ProtectiveCollar-Fall-2026.ipynb

Review on your own:

CHEME-5660-L11b-Example-CoveredCallFund-SPY-Fall-2026.ipynb

CHEME-5660-L11b-Derivation-NetDelta-Fall-2026.ipynb

Company Profile: J.P. Morgan Asset Management

J.P. Morgan Asset Management: the investment management business of JPMorganChase. It runs two large covered call funds, JEPI (S&P 500) and JEPQ (Nasdaq-100):

- Shares plus sold options: each fund owns U.S. large-cap stocks and sells calls on its index, seeking monthly income from the premiums.
- Equity-linked notes: the calls sit inside notes issued by banks; if a bank fails to pay, the fund can lose money.
- Size: about 45 (JEPI) and 44 (JEPQ) billion USD on September 30, 2026.
- Volatility: 7.78% (JEPI) vs 12.42% (S&P 500) per year.

Connection: the calls are on the index, not on the stocks the fund owns, so the two can move apart.

A Closer Model: REX and FEPI

The REX FANG & Innovation Equity Premium Income ETF (FEPI) runs the covered call from today's lecture:

- Shares plus calls on those shares: it owns 15 large technology stocks, AMD among them, and sells slightly out-of-the-money calls on each, about one contract per 100 shares.
- Size: about 0.65 billion USD on July 31, 2026 (annual report to the SEC).

Connection: over the year ending July 31, 2026, FEPI trailed its index because the sold calls gave up gains when technology stocks rose.

Covered Calls

A covered call owns shares and sells a call on them, collecting the premium. Investors use it when they expect the share price to stay flat or rise slightly.

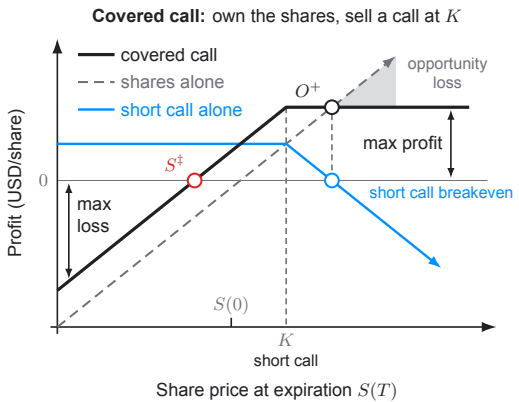
Naked calls: selling a call without owning the shares carries unlimited risk if the share price rises. The shares in a covered call cover that obligation.

Besides the funds in the company profiles, covered call ETFs such as QYLD run this strategy for you.

Video: SMB Capital, Simple & Low Risk Investing Strategy

Covered Call: Shares Plus a Short Call

We buy shares at $S(0)$ and sell one call at the strike K per m shares. Per share, we pay $S(0) - \mathcal{P}_c$ today:



Covered Call: Profit at Expiration

Buy shares at $S(0)$ and sell one call per m shares, with strike K, premium $\mathcal{P}_c$, and expiration T. The profit at expiration (USD/share) is:

$$\begin{aligned}\hat{P}(S(T)) &= S(T) - (S(T)-K)^+ - \mathcal{D}_{T,0}(g_y)\cdot(S(0)-\mathcal{P}_c) \\ &= \underbrace{\big(S(T) - \mathcal{D}_{T,0}(g_y)\cdot S(0)\big)}_{\text{shares}} + \underbrace{\mathcal{D}_{T,0}(g_y)\cdot\mathcal{P}_c - (S(T)-K)^+}_{\text{short call}}\end{aligned}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor; for n contracts, multiply by mn. With no dividends and no early exercise:

$$\hat{P}(S(T)) = \begin{cases} \overbrace{K - \mathcal{D}_{T,0}(g_y)\cdot(S(0)-\mathcal{P}_c)}^{\text{realized gain}}, & \text{if } S(T) > K \text{ (call exercised)} \\ \mathcal{D}_{T,0}(g_y)\cdot\mathcal{P}_c + \underbrace{\big(S(T) - \mathcal{D}_{T,0}(g_y)\cdot S(0)\big)}_{\substack{\text{unrealized gain} \\ \text{(no share sale)}}}, & \text{if } S(T) \le K \text{ (call expires worthless)} \end{cases}$$

Covered Call: Breakeven, Maximum Profit, and Maximum Loss

The maximum profit, breakeven $S^{\ddagger}$, maximum loss, and opportunity point O^{+} are given below. Maximum losses are listed as positive amounts (USD/share):

Maximum profit	$K - \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$ for $S(T) \geq K$
Breakeven	$S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$
Maximum loss	$\mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$ at $S(T) = 0$
Opportunity point	$O^{+} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$, the breakeven of the short call alone

Compared with owning the shares alone, the premium lowers the breakeven, and the call gives up the gains above the strike. If we bought the shares earlier at a cost basis $S^{\dagger}$, the realized gain on the shares at exercise is $K - S^{\dagger}$.

Example: CHEME-5660-L11b-Example-AnalysisCoveredCall-Fall-2026.ipynb **Review on your own:** CHEME-5660-L11b-Example-CoveredCallFund-SPY-Fall-2026.ipynb

Protective Collars

A protective collar adds a long put to a covered call: own the shares, buy a put at K_1 below the share price, and sell a call at $K_2 > K_1$ above it, same expiration T. Per m shares, with premiums $\mathcal{P}_1$ (put) and $\mathcal{P}_2$ (call), the profit at expiration (USD/share) is:

$$\hat{P}(S(T)) = S(T) + \underbrace{(K_1 - S(T))^+}_{\text{long put}} - \underbrace{(S(T) - K_2)^+}_{\text{short call}} - \underbrace{\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)}_{\text{today's cost, grown}}$$

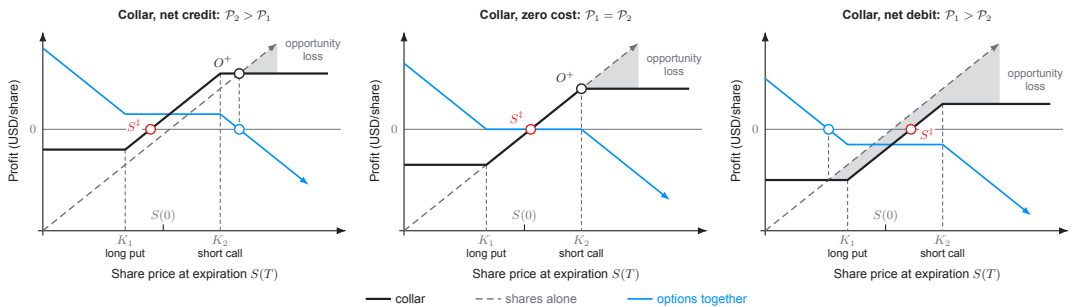
The put sets a floor under the shares, the call caps them, and the call premium pays for part of the put, all of it, or more (a net credit).

With no dividends and no early exercise, at most one option pays at expiration:

$$\hat{P}(S(T)) = \begin{cases} K_1 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2), & \text{if } S(T) < K_1 \text{ (put exercised)} \\ S(T) - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2), & \text{if } K_1 \le S(T) \le K_2 \text{ (both expire worthless)} \\ K_2 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2), & \text{if } S(T) > K_2 \text{ (call exercised)} \end{cases}$$

Protective Collar: Net Credit, Zero Cost, and Net Debit

The collar (black) against the shares alone (gray) and the two options together (blue). The shading marks where the shares alone earn more:



Protective Collar: Breakeven, Maximum Profit, and Maximum Loss

The maximum profit, breakeven $S^{\ddagger}$, maximum loss, and opportunity point O^{+} are given below. Maximum losses are listed as positive amounts (USD/share):

Maximum profit	$K_2 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$
Breakeven	$S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$, if $K_1 \leq S^{\ddagger} \leq K_2$
Maximum loss	$\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2) - K_1$ for $S(T) \leq K_1$ [a]
Opportunity point	$O^{+} = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ [b]

[a] If $\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2) > K_1$; otherwise none. [b] For a net debit, the shares alone earn more for any $S(T) > K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$.

When $\mathcal{P}_1 = \mathcal{P}_2$, the call premium pays for the whole put: a zero-cost collar. Compared with the covered call, the put limits the loss, and it costs some of the call premium.

TJUL: A Collar Inside a Fund

The Innovator Equity Defined Protection ETF (TJUL) runs a collar on SPY for you, with a two-year expiration; the current period runs from July 1, 2025 to June 30, 2027. On July 31, 2026, it held three options on SPY, all expiring June 30, 2027:

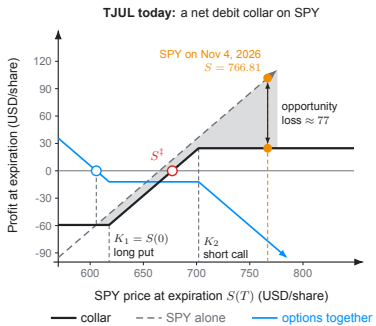
- Long call, strike 11.12: so deep in the money that it moves almost one for one with SPY; the fund holds it instead of the shares (no dividends).
- Long put, strike 617.85: SPY started at 617.85, so this put protects against any fall in SPY below that price.
- Short call, strike 701.94 ($= 617.85 \times 1.1361$): caps the gain, as in a covered call.

The fund's goal: held for the whole two years, its share price rises as much as the price of SPY, up to 13.61%, and does not fall if SPY falls (before fees).

Example: CHEME-5660-L11b-Example-StandardListed-SPY-ProtectiveCollar-Fall-2026.ipynb **Source:** SEC N-PORT, July 31, 2026

TJUL Today

TJUL's July 31, 2025 report on EDGAR, the SEC's free database of filings: $\mathcal{P}_1 \approx 45.46$, $\mathcal{P}_2 \approx 34.23$ USD/share, a net debit:



Source: SEC N-PORT, July 31, 2025

Delta of Covered Calls and Collars

Previously, we used Delta as an estimate of the probability of exercise. Delta also measures how much a position gains or loses when the share price moves. The net Delta is the shares plus the weighted Deltas of the option legs.

A covered call holds mn shares and is short n calls with Delta Δ_c; a collar adds n long puts with Delta Δ_p (put at K_1, call at K_2):

$$\Delta_{\text{net}}(S,t) = mn\,(1 - \Delta_c(S,t)) \qquad \text{covered call}$$

$$\Delta_{\text{net}}(S,t) = mn\,\big(1 + \Delta_p(S,t) - \Delta_c(S,t)\big) \qquad \text{collar}$$

Since $0 \le \Delta_c \le 1$, the covered call's net Delta is never negative. Since $-1 \le \Delta_p \le 0$, the put and the call can only lower the collar's net Delta from mn.

Derivation: CHEME-5660-L11b-Derivation-NetDelta-Fall-2026.ipynb

Summary

We analyzed covered calls and protective collars, two strategies that combine shares with options, and computed the net Delta of each position.

Key takeaways

- Covered calls: the call caps the profit at the strike, and the premium lowers the breakeven.
- Protective collars: the put sets a floor and the call a cap; in between, the position moves with the shares.
- Net Delta: the shares plus the weighted Deltas of the option legs; a collar's put and call can only lower it from that of the shares.

These strategies trade some of the upside of the shares for income or for protection.